
SPARSITY AND SUPERPOSITION IN MIXTURE OF EXPERTS

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Anonymous submission

ABSTRACT

Mixture of Experts (MoE) models have become central to scaling large language models, yet their mechanistic differences from dense networks remain poorly understood. Previous work has explored how dense models use *superposition* to represent more features than dimensions, and how superposition is a function of feature sparsity and feature importance. We show that MoE models cannot be explained mechanistically through the same lens, and we quantify the gap directly. Across every configuration we test, neither feature sparsity nor feature importance induces the discontinuous phase changes that dense models reliably exhibit; instead, the single variable that consistently tracks representational strategy is network sparsity (the ratio of active to total experts). Using new metrics for measuring superposition across experts, we find a clear and monotonic pattern: the more experts a MoE has, the less superposition it carries, so that models with greater network sparsity are systematically more *monosemantic*. Critically, this interpretability gain comes at a small and shrinking cost—MoEs trail the dense model in reconstruction loss by only ~ 0.03 – 0.08 at matched total parameters, and this margin narrows steadily as the expert count grows. Building on this, we propose a new definition of expert specialization based on monosemantic feature representation rather than load balancing, and show quantitatively that experts organize around coherent feature combinations: under k-hot initialization, *100%* of the features an expert represents monosemantically are occupied by that same expert. Together these results indicate that network sparsity in MoEs buys markedly more interpretable models for a modest and diminishing performance cost, directly challenging the common assumption that interpretability and capability are fundamentally at odds.

1 INTRODUCTION

Mixture of Experts (MoEs) have become prevalent in state-of-the-art language models, such as Qwen3, Mixtral, and Gemini (Yang et al., 2025a; Jiang et al., 2024; Google DeepMind, 2025), primarily for their documented computational efficiency and consistent performance gains (Shazeer et al., 2017; Fedus et al., 2022). Subsequent work improves routing (e.g., Expert-Choice routing) and training stability/transfer (e.g., ST-MoE) (Zhou et al., 2022; Zoph et al., 2022). Theoretical and empirical results further show that learnable routers can discover latent cluster structure in data, providing insight for why experts specialize (Dikkala et al., 2023). Yet despite this widespread adoption and a growing body of quantitative results on their efficiency, MoEs remain poorly understood from a mechanistic interpretability perspective—we lack the kind of controlled, measured comparison against dense models that would explain *why* they behave differently.

Interpretability-oriented approaches have sought to make expert behavior more transparent. Yang et al. (2025b) proposes MOE-X, which encourages sparsity-aware routing and uses wide, ReLU-based experts to reduce polysemanticity. Park et al. (2025) introduce MONET, scaling the number of experts to enable capability editing via expert activation. Yet these works largely rely on architectural changes to *impose* interpretability, and none quantify whether the MoE structure itself already reduces superposition relative to a matched dense

baseline. We still lack a measured, mechanistic account of how MoEs represent features, how the expert count affects superposition, and whether experts naturally specialize without extra regularization. Mu & Lin (2025) survey MoE research and single out mechanistic interpretability as a key open challenge.

A fundamental challenge in interpreting neural networks is the phenomenon of superposition: when models represent more features than they have dimensions. This trade-off is quantifiable—networks pack many sparse features into fewer neurons, driving the features-per-dimension ratio above one, at the direct cost of making individual neurons polysemantic and difficult to interpret.

MoE architectures introduce a new axis to this problem: network sparsity. Unlike dense models that activate all neurons regardless of input, MoEs activate only a fraction of their total parameters (Shazeer et al., 2017). Where dense models exploit feature sparsity by packing many sparse features into shared neurons, MoEs can afford to be more selective, potentially dedicating entire experts to specific feature combinations. As we show, this is not a marginal difference but the dominant factor: it is network sparsity, not feature sparsity or importance, that most cleanly predicts how much superposition a model carries.

We investigate whether (1) MoEs exhibit less superposition than their dense counterparts, (2) there is a discrete phase change in the amount of superposition of a particular feature in MoE experts across its relative importance and overall feature sparsity, as seen in dense models, and (3) we can understand expert specialization through the lens of feature representation rather than just load balancing.

We explore these questions using simple models that extend Elhage et al. (2022)’s framework to MoEs, and we ground each claim in a direct, matched comparison against a dense baseline. Our key contributions are as follows: (1) across all three architectures we study, dense single-expert models reproduce the sharp, discontinuous phase changes of prior work, whereas every MoE we train replaces these discontinuities with smooth transitions—an effect that strengthens monotonically with network sparsity; (2) at equivalent active *and* total parameters, MoEs consistently exhibit greater monosemanticity (less superposition) than the dense model, with the features-per-dimension ratio falling from the dense model’s superposition regime toward the monosemantic limit of 1.0 as experts are added, and individual experts representing features more cleanly; (3) we propose an interpretability-focused definition of expert specialization based on monosemantic feature representation, and show quantitatively—across 1000 experts per initialization scheme—that experts organize around coherent feature combinations rather than arbitrary load balancing, with expert usage rising sharply and predictably whenever an expert’s monosemantic features are active.

2 RELATED WORK

Superposition and Feature Representations. The Linear Representation Hypothesis suggests networks represent concepts as directions in activation space (Park et al., 2024), yet the number of interpretable features often exceeds the available dimensions. Elhage et al. (2022) formalized this phenomenon as superposition, demonstrating quantitatively that dense models rely on non-orthogonal feature packing to maximize capacity at the cost of polysemanticity. While superposition is theoretically efficient (Scherlis et al., 2025), it necessitates complex post-hoc disentanglement methods, such as Sparse Autoencoders, to recover monosemantic features (Bricken et al., 2023). Recent work has examined how data correlations shape superposition (Prieto et al., 2025), and how interference patterns emerge and can be mitigated (Gurnee et al., 2023). We extend this line of inquiry to MoE architectures and, using the same feature-norm and interference measurements, show that the controlling variable shifts: network sparsity—rather than feature sparsity alone—governs representational strategy.

Interpretability of MoEs. While MoEs have become the standard for scaling large language models (Shazeer et al., 2017; Fedus et al., 2022; Jiang et al., 2024), mechanistic understanding of their internal representations lags behind their dense counterparts. Existing analysis largely focuses on macroscopic behaviors, such as routing stability (Zoph

et al., 2022), expert choice statistics (Zhou et al., 2022), or latent cluster discovery (Dikkala et al., 2023). More recent interpretability-focused approaches attempt to force specialization through architectural constraints, such as predefined concept routing (Yang et al., 2025b) or scaling expert counts to match vocabulary sizes (Park et al., 2025). However, these approaches often prioritize capability editing over explaining intrinsic feature geometry, and rarely quantify the interpretability benefit against a matched dense baseline. We close this gap with a controlled, measured comparison: holding parameters fixed, we show that MoEs are consistently more monosemantic than dense models, and we ground a feature-based definition of expert specialization in usage statistics aggregated over thousands of experts.

3 BACKGROUND

A primary focus of Mechanistic Interpretability is to reverse engineer neural networks; one method is to decompose model representations into a set of human interpretable concepts named ‘features’. These features are often assumed to be linear, that is, any hidden state h can be described as

$$h = \sum_{i \in F} \alpha_i \vec{f}_i + \vec{b}$$

where $\vec{f}_i$ is the direction corresponding to feature i , α_i is the activation strength of this feature (roughly, the degree to which feature i is present in the input), and F is the set of all represented features.

The *superposition hypothesis* contends that models can represent far more features than dimensions, i.e. $|F| > m$ for $h \in \mathbb{R}^m$ (Elhage et al., 2022). To fit many more features than dimensions into a latent space, the feature vectors in F must be packed so that they are not all orthogonal. When two features have interference $\langle \vec{f}_i, \vec{f}_j \rangle \neq 0$, we say they are in *superposition*. This packing is acceptable so long as features are sparsely active (a characteristic of most online text data), but it carries a measurable interpretability cost: α_i then contains spurious activations unrelated to feature i being present in the input. The degree of this cost is exactly what our metrics quantify and what we compare across dense and MoE models.

Monosemanticity is a characteristic of individual neurons, where a neuron’s activation cleanly corresponds with a single α_i (i.e., features are basis-aligned). When features are orthogonal (not in superposition) but not basis-aligned, neurons remain *polysemantic* even though feature interference is minimal. In this paper, we focus on reducing superposition rather than enforcing basis-alignment; this is the quantity our interference and features-per-dimension metrics track. For brevity, when we describe models, experts, or features as “more monosemantic,” we mean they display measurably less superposition.

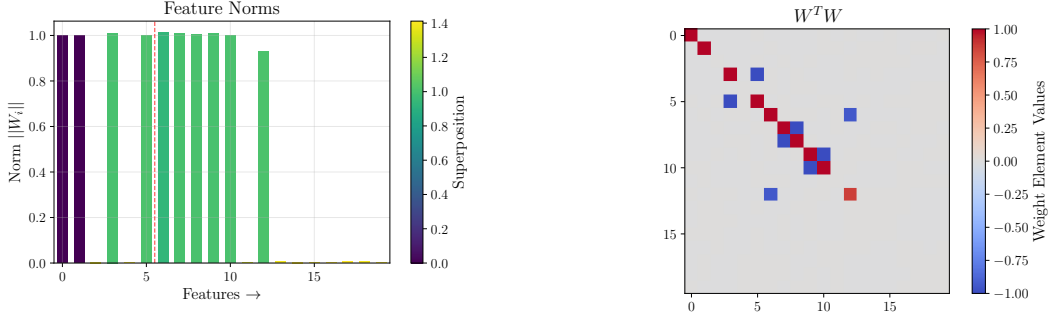
4 DEMONSTRATING SUPERPOSITION

MoEs are often conceptualized as compositions of dense models, where each expert behaves like an independent dense network. Whether experts actually represent features the way dense models do, however, is an empirical question. We answer it with a direct, parameter-matched comparison of superposition between MoEs and dense models, measuring feature norms and interference under identical input distributions.

4.1 EXPERIMENTAL SETUP

Our goal is to explore how a MoE can project a high-dimensional vector, $x \in \mathbb{R}^n$ into lower-dimensional expert representations, $h \in \mathbb{R}^m$ and then accurately recover it. This extends the framework of Elhage et al. (2022) to the MoE setting.

The input distribution. The input vector x represents the activations of an idealized, disentangled model, where each dimension x_i corresponds to a distinct, independent feature—effectively, a perfectly neuron-aligned and monosemantic representation. We take $x_i \sim U(0, 1)$, except that for a given sparsity $S \in [0, 1)$ we set $P(x_i = 0) = S$. Not all



(a) Norm of each feature’s weight vector $\|W_i\|$, with colors indicating superposition status (green for features in superposition, purple for monosemantic features).

(b) $W^T W$ matrix where each cell represents $(\hat{W}_i \cdot W_j)$, revealing interference patterns between features.

Figure 1: **Dense baseline.** Feature representation and superposition in a dense model with $n = 20$ features and $m = 6$ hidden dimensions, with importance $I = 0.7^i$ and uniform feature density $(1 - S) = 0.1$. Superposition (color) is given by $\sum_j (\hat{W}_i \cdot W_j)^2$. Note the fully off-diagonal interference structure in (b): any feature can interfere with any other, the hallmark of dense-model superposition that the matched MoE in Figure 2 avoids.

features contribute equally to the loss. To model that features vary in utility, we assign each feature x_i a scalar importance I_i which weights the reconstruction loss. Additionally, to isolate and study phase transitions for a single feature, we scale the magnitude of the last feature by a factor $r \in \mathbb{R}^+$, such that $I = (1, 1, \dots, r)$. These two knobs, I_i and r , are precisely the axes along which dense models show phase changes; varying them lets us test—on the same axes—whether MoE experts respond the same way.

Model Architecture. The MoE consists of E experts, where each expert e is parameterized by a weight matrix $W^e \in \mathbb{R}^{m \times n}$ and a bias $b^e \in \mathbb{R}^n$. Inputs are assigned to the top- k experts via a learned router $g(x) = \text{softmax}(W^r x)$ where $W^r \in \mathbb{R}^{E \times n}$. Each active expert projects the input to a lower-dimensional hidden state $h^e = W^e x$ and generates a reconstruction $\hat{x}^e = \text{ReLU}((W^e)^T h^e + b^e)$ ¹. The final output is the weighted sum of the active experts: $x' = \sum_{e \in \text{top-}k} w_e \hat{x}^e$ where w_e are the renormalized gating weights.

We train our models with an L2 reconstruction loss weighted by feature importances I_i , given by $\mathcal{L} = \sum_x \sum_i I_i (x_i - x'_i)^2$. In Section 4, to prevent expert collapse, we add the standard auxiliary load balancing loss (Fedus et al., 2022), defined as $\mathcal{L}_{\text{aux}} = \alpha N \sum_{e=1}^N f_e P_e$. Here, N is the total number of experts, f_e is the fraction of samples in a batch routed to expert e , P_e is the average gating probability assigned to expert e across the batch, and $\alpha = 0.01$ controls the penalty strength. We deliberately omit auxiliary load balancing in Section 4, so that any measured difference in superposition reflects the intrinsic architectural bias of the MoE rather than a regularizer—keeping the dense-versus-MoE comparison as clean as possible.

4.2 MEASURING FEATURE CAPACITY

To analyze feature representations across architectures, we compared two fundamental properties of the features: *representation strength* and *interference* with other features. We measured the norm of a feature weight vector in an expert e , $\|W_i^e\|$, which quantifies how strongly feature i is represented within expert e : $\|W_i^e\| \approx 1$ when the feature is fully repre-

¹The ReLU at the output ensures non-negative reconstructions, which matches our input distribution where $x_i \in [0, 1]$ when nonzero. Furthermore, the off-diagonal terms in $(W^e)^T W^e$ create negative interference and ReLU suppresses these negative components. When features are sparse, negative interference becomes effectively “free” as it is filtered to zero, incentivizing model configurations with negative off-diagonal terms (e.g., antipodal pairs).

sented and ≈ 0 when it is not learned. We visualize interference between features in expert e using the Gram matrix $W^\top W$, whose off-diagonal elements give the pairwise interference.

The comparison is direct and matched. As shown in Figures 1a and 2a, at equal total parameters ($m_{\text{dense}} = \sum m_{\text{experts}} = 6$) the dense model and the MoE learn a comparable number of features (10 vs. 8) with similar norms—so the two architectures reach the same representational capacity. The difference is entirely in *how* that capacity is packed. The MoE experts show only mild local superposition (e.g., Expert 0 in Fig. 2b), whereas the dense model’s interference is global: any feature can interfere with any other (Figure 1b). The MoE instead enforces a block-diagonal structure in which features routed to different experts have *exactly zero* interference. In short, at matched capacity the MoE partitions the feature space and confines interference locally, sharply reducing its global scope.

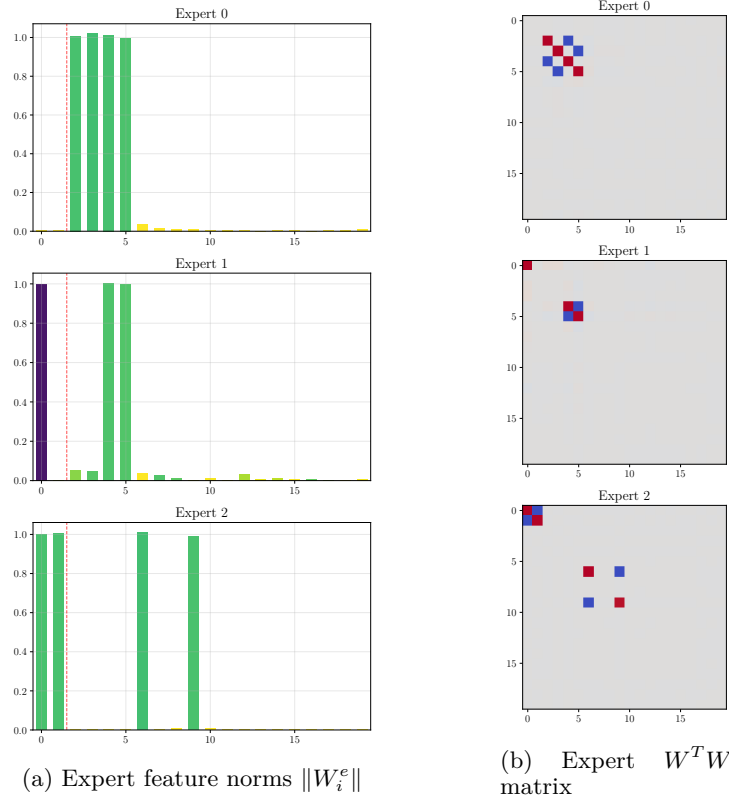


Figure 2: Matched MoE. Feature representation and superposition in a MoE with $n = 20$ features, 3 total experts, and $m = 2$ hidden dimensions per expert (top- $k = 1$ routing), with importance $I = 0.7^i$ and feature density $1 - S = 0.1$. Total parameters match the dense model of Figure 1 ($3 \times 2 = 6$). The per-expert $W^\top W$ matrix (b) is nearly block-diagonal: features routed to different experts carry zero interference, in contrast to the dense model’s fully off-diagonal structure.

Expert Feature Dimensionality. We want to understand how MoEs allocate their limited representation capacity differently from the dense model. We measured feature dimensionality, which represents the “fraction of a dimension” that a specific feature gets in a model (Elhage et al., 2022). For a feature i , we define its dimensionality in expert e by

$$D_i^e = \frac{\|W_i^e\|^2}{\sum_j \left(\hat{W}_i^e \cdot W_j^e \right)^2} \quad (1)$$

D_i^e is bounded between zero (not learned) and one (monosemantic). The total capacity for a MoE can thus be defined as $D = \sum_e \sum_{i=1}^n D_i^e$.

Efficient Packing. When the features are “efficiently packed” in a model’s representation space, the dimensionalities of all features sum to the number of embedding dimensions, i.e. $\sum_{i=1}^n D_i^e \approx m$ (Cohen et al., 2014; Scherlis et al., 2025; Elhage et al., 2022). For a MoE, the relation becomes $\sum_e \sum_{i=1}^n D_i^e \approx E \cdot m$. Empirically, both dense and MoE models satisfy this constraint. This is an important control: it establishes that MoEs are not simply wasting capacity, but pack features just as efficiently as dense models at the same total parameters. Any difference we observe in superposition therefore reflects a genuine difference in representational *strategy*, not in efficiency.

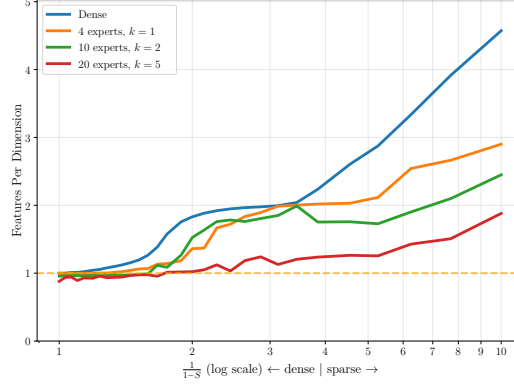


Figure 3: **More experts, less superposition.** Features per dimension versus inverse feature density ($\frac{1}{1-S}$) for **dense** and **MoE** architectures with uniform feature importance ($I_i = 1.0$), all at equal total parameters and similar k/E . The ordering is consistent across every density level: the **dense** model ($n = 100$, $m = 20$) carries the most superposition, and this falls monotonically as experts are added—4 experts, $m = 5$, $k = 1$ (orange); 10 experts, $m = 2$, $k = 2$ (green); 20 experts, $m = 1$, $k = 5$ (red). The dashed line at 1.0 marks the monosemantic limit; the MoE curves sit at or near it while the dense curve rises well above, quantifying how much extra superposition the dense model uses.

Features per Dimension & Network Sparsity. Since both dense and MoE models “efficiently pack” features in their representation space, we compared the differences in number of features per dimension across the models. This allowed us to exactly measure superposition in both models and how the number of experts in a MoE affects superposition for different feature sparsities. If the features per dimension is greater than one, then the features are in superposition since the model is representing more features than there are dimensions. We define features per dimension for a MoE by

$$\frac{1}{k} \sum_{e=1}^E p_e \frac{\|W^e\|_F^2}{m} \quad (2)$$

where $\|W\|_F^2$ is the Frobenius norm and p_e is the expected probability that expert e is used across a batch of input samples, i.e. the average renormalized gating weight after top- k routing.

At *equal* total parameters, the pattern in Figure 3 is unambiguous: the dense model has the highest features-per-dimension at every feature density we test, i.e. the most superposition. The dense model uses superposition to represent a greater total number of features ($n_{\text{learned}} > m_{\text{total}}$), whereas the MoE holds its representation at the monosemantic limit ($n_{\text{learned}} \approx m_{\text{total}}$). The trend across the MoE curves is equally clean and monotonic: increasing the total number of experts—while holding total parameters and the ratio k/E roughly fixed—drives features-per-dimension steadily downward toward 1.0. *The more experts, the less superposition.* Features become measurably more monosemantic as the expert count grows.

This interpretability gain carries only a small, quantifiable performance cost. The dense model’s extra superposition buys it a consistently lower reconstruction loss than the MoEs (Figure 6, Appendix A.1), but the gap is modest—only ~ 0.03 – 0.08 at any given sparsity—and it narrows steadily as experts are added, with the largest MoEs reaching loss essentially comparable to the dense model. In other words, MoEs recover most of the dense model’s performance while shedding most of its superposition. See Appendix A.2 for a theoretical intuition.

5 PHASE CHANGE

Having established that MoEs and dense models learn a similar number of features but that MoEs distribute them with less interference, we now ask what drives this allocation. The prior section shows network sparsity reshapes *how* features are allocated rather than how many are learned; here we test whether the input-distribution properties that govern dense models—feature sparsity and importance—still induce the physics-inspired *phase changes* that are the signature of dense superposition, or whether that signature disappears in MoEs.

Models have a finite way of representing features; each feature may be ignored, superimposed, or monosemantic. Phase change is the observation that sometimes there are discrete boundaries between regions, which are functions of feature sparsity and relative importance.

Dense toy models exhibit discontinuous ‘phase changes’ between internal feature representations (Elhage et al., 2022). Varying the sparsity and relative importance of features in the input distribution elicits distinct regimes—for example, more feature sparsity encourages greater superposition. When we plot the same expert-level phase diagrams for MoEs, the contrast with the dense case is stark: MoE experts employ visibly different representational strategies, as the sections below quantify.

Setup. We follow the same setup as Section 4.1, except that we add a load balancing loss to avoid specialization collapse—when experts and routers fall into local minima where certain experts are entirely ignored—in certain setups. We sweep three architectures, all with one active expert ($k = 1$): (A) $n = 2, m = 1$; (B) $n = 3, m = 1$; and (C) $n = 3, m = 2$. For each, we report the expert-specific phase diagram across feature sparsity and last-feature relative importance, and vary network sparsity by increasing the expert count E up to the number of input feature dimensions n . This lets us read off, within a single architecture, exactly how the phase-change signature changes as network sparsity increases.

In this section we fix *active* parameters rather than total parameters, so that $m_{dense} = km_{moe}$ (ignoring router parameters). This is the controlled comparison we want: it isolates the effect of network sparsity, since otherwise any observed difference could be attributed to a change in the number of active parameters rather than to the MoE structure itself. Coincidentally, 4.C.1/1 has the same number of active parameters as 4.B.X/2; but the latter has only one hidden dimension ($m = 1$) to encode the same $n = 3$ input features as the former’s two ($m = 2$), which makes superposition a less useful lens for understanding specialization there.

Results and Takeaways. The comparison is clear-cut. In every single-expert (dense) case, we observed a sharp phase change (Figure 4.X.1/1), reproducing Elhage et al. (2022). The moment we increased the total number of experts, the discrete phase changes disappeared—this held across all three architectures. Some $E = 2$ experts still resemble their dense counterparts (Figure 4.X.2/2), but even these replace the discontinuity with a continuous transition; and in each pair, one expert consistently becomes more monosemantic, specializing in the highest-importance feature. The experts that do *not* resemble the dense case are, without exception, markedly bluer—lower superposition scores, more monosemantic representations. This is exactly the direction predicted by the previous section: across both experiments, MoEs land at systematically lower superposition than their dense counterparts.

The per-architecture patterns reinforce the same story. For $n = 2, m = 1$ (Figure 4.A), the dense model drops the last feature at low feature sparsity, whereas the matched MoE preserves it substantially more, because the added expert capacity gives it room to. With three input dimensions (Figure 4.B) this advantage disappears: the experts already super-

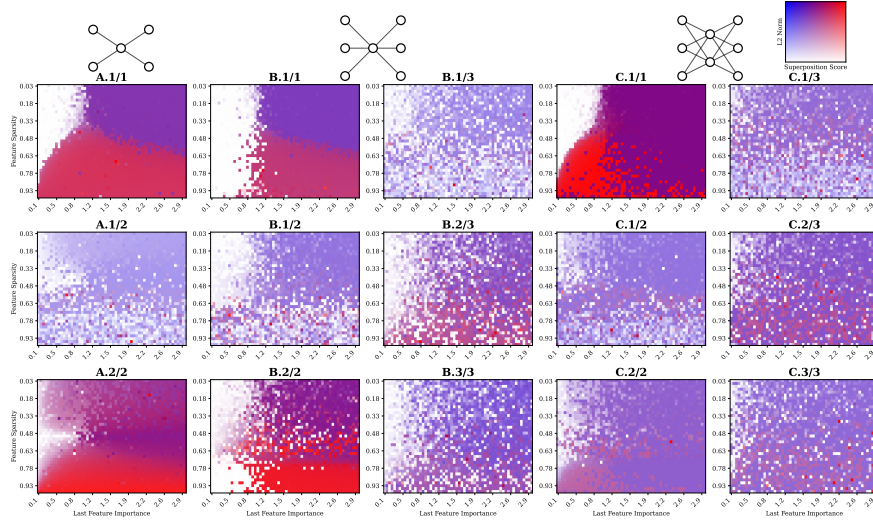


Figure 4: **Phase changes vanish as network sparsity rises.** For a particular expert and input feature, we decode how the feature is embedded in the hidden dimension—ignored (white), monosemantic (blue-purple), or superimposed (red). We plot joint feature norm ($\|W_n\|^2$) and superposition score ($\sum_{j < n} (\hat{W}_n \cdot W_j)^2$) for the last feature across feature sparsity $S \in [0.1, 1]$ and relative last-feature importance $I_n \in [0.1, 3]$, where the subscript n denotes the last of n total features. Low L2 norm is white (the feature is ignored); otherwise, low superposition is blue-purple (monosemantic) and high superposition is red. For each cell we train ten models and keep the lowest-loss one; a load balancing loss is used throughout this section. Cell (i, j) in subfigure X.e/E is expert e of E total experts, architecture X, importance $I_n = i$, sparsity $S = j$; X.1/1 is the dense baseline. Reading each row left to right (increasing E), the sharp dense-model boundaries dissolve into smooth, predominantly blue-purple transitions—the same monosemantic shift quantified in Figure 3.

impose the other two features, leaving no space for a third within a single hidden dimension, so in the high-sparsity regime one hidden dimension with superposition simply cannot hold all features. Even here, though, no sharp phase change appears—the only exception is the 0.5–0.7 feature-sparsity band in 4.B.2/2, which is mostly discrete but mixed. For $m = 2$, the dense model’s white region (Figure 4.C.1/1, mid- to low-sparsity, where the last feature is relatively less important) ignores that feature entirely. As network sparsity increases across the remaining Figure 4.C panels, the models represent the last feature with strictly increasing L2 magnitude ($\|W_3^1\| < \|W_3^2\| < \|W_3^3\|$). That ordering is the point: dimensionality in the low-importance region rises monotonically with network sparsity, mirroring the features-per-dimension trend of Figure 3.

In a consistent window of feature sparsity, roughly 0.48 to 0.7, three of the two-expert panels (Figures 4.B.2/2, 4.C.1/2, and 4.C.2/2) show a heavy mix of polysemantic, monosemantic, and ignored representations. This points to a genuine middle ground in MoEs where polysemantic and monosemantic strategies achieve comparable loss, so the model does not commit cleanly to either—unlike the decisive behavior it shows in the low- and high-sparsity regimes. No dense model exhibits this window, which is further evidence that MoEs learn qualitatively different representational strategies.

Conclusion. These experiments are all top- $k = 1$, so only one expert is active at a time—the setting that most resembles a dense model. Yet even in the minimal $E = 2$ case we already see substantially different behavior, including when a single expert’s superposition capacity is sufficient to represent all features. The evidence therefore weighs against treating a MoE as an aggregation of dense models: the router, by letting each expert observe only a subset of the feature domain, materially changes what and how the experts learn.

6 EXPERT SPECIALIZATION

Having established that MoEs carry less superposition, we now examine how these more monosemantic features are organized within experts, and what that organization reveals about specialization.

Expert specialization in MoEs is traditionally framed in terms of load balancing across all inputs (Chaudhari et al., 2025). That framing misses the intuitive notion of specialization: an expert should be invoked precisely when the concepts it specializes in are present in the input.

We instead define an expert as specialized if it *occupies* certain feature directions in the input space *and* represents those features relatively monosemantically. We show empirically that these two conditions are strongly correlated, and that their co-occurrence in turn encourages balanced load across experts—turning load balancing into a consequence of specialization rather than its definition.

Because we fix $k = 1$, the feature space is partitioned into convex cone regions (see Appendix A.3), with each region routed to a particular expert. By definition, this means $\forall s > 0, \vec{x} \in C \rightarrow s\vec{x} \in C$, where C is the set of points contained within the cone and s is any positive scalar. If a particular feature vector $\vec{x}$ is routed to an expert, then all $s\vec{x}$ are routed to that same expert. In this case, we say that feature x is contained within expert e , and as such expert e *occupies* x .

We empirically find that small models which spread the input space across more experts achieve lower loss (Appendix A.4). This raises the question of whether allocating the input space to particular experts implies anything measurable about those experts. Our definition predicts that it implies *monosemanticity*—a correlation we confirm quantitatively below and show holds even for larger toy models (e.g., $m = 10$).

In models with $n > 2$, we test two directions of this correlation. First, whether initializing experts to occupy specific features in the input space causes those experts to represent said features more monosemantically. Second, and conversely, whether—for the features an expert chooses to represent monosemantically—that expert also occupies those features in the input space.

The three schemes give a graded, interpretable result. Under diagonal initialization, each expert monosemantically represents the single feature it initially occupied—and only that feature (Figure 5a). Under ordered k-hot initialization, the first expert monosemantically represents four of the five features it occupied (Figure 5b); the remaining experts, assigned to lower-importance features, represented neither those features nor the five most important features they were not assigned. When we break the importance ordering and randomize which features each expert occupies, every expert monosemantically represents only the most important feature(s) it was assigned (Figure 5c). The common thread is that monosemantic representation is reserved for the occupied *and* important features.

These results establish a strong correlation between the features initially routed to an expert and the features that expert ultimately represents monosemantically. Moreover, experts reserve monosemantic representation for *important* features—whether we hand an expert a single important feature outright, or give it a set and let it select the most important one to represent cleanly on its own.

The correlation persists even in the hardest case. Under uniform feature importance, experts place all features in superposition with elevated interference; yet the features an expert initially occupies remain *relatively* more monosemantic, achieving superposition scores on average 1.03 standard deviations below that expert’s mean. In other words, occupancy predicts relative monosemanticity by roughly one standard deviation even when no feature is privileged by importance.

We next tested the converse direction: whether experts that represent certain features monosemantically also occupy those features in the input. We measure expert usage in two conditions—when the monosemantic feature(s) are *one of many* active features, and when they are the *only* active feature (the latter directly estimates the probability that the expert

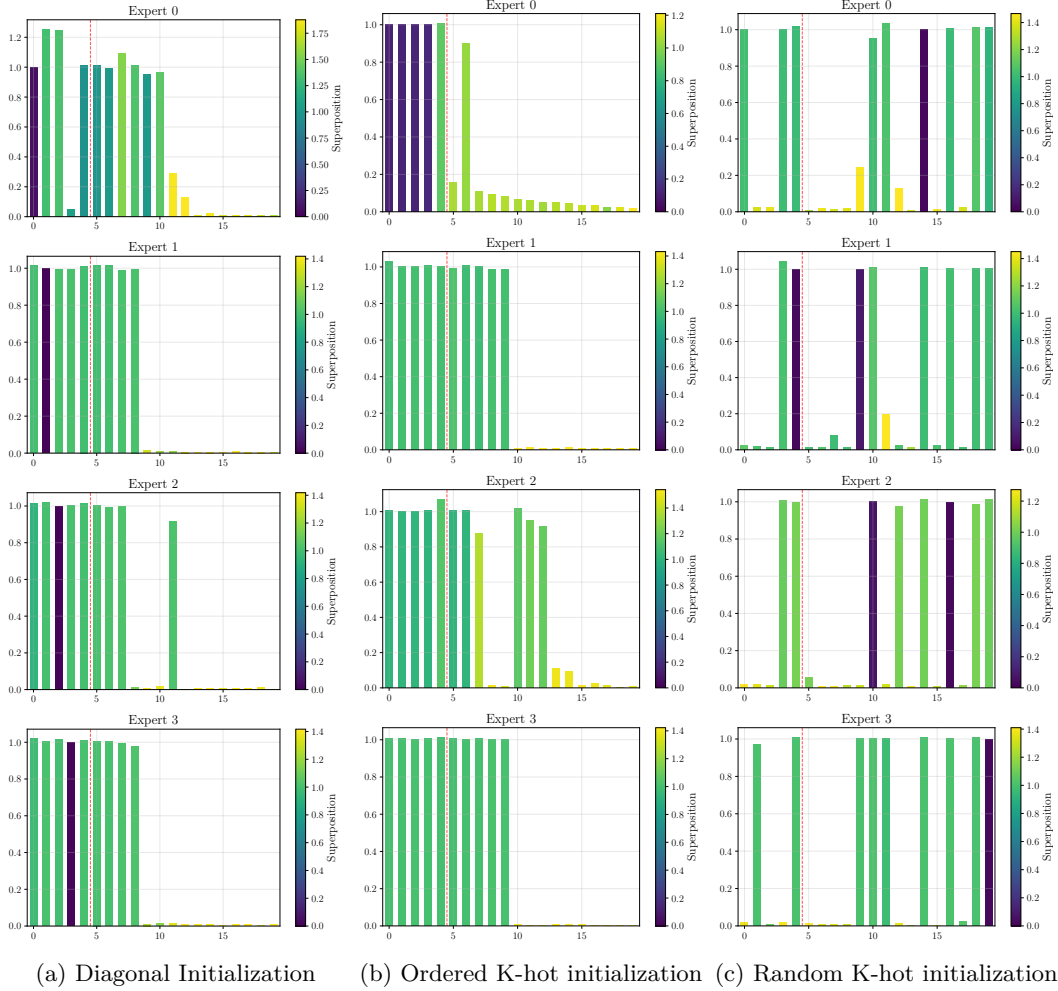


Figure 5: Initialization controls which features an expert represents monosemantically. Expert feature norms $\|W_i^{(e)}\|$ and superposition (color) across three initialization schemes, all with $n = 20, m = 5, E = 4, S = 0.1$. **(a)** Diagonal initialization ($W_i^r = \hat{e}_i$), importance decreasing exponentially from feature 1 to 20: each expert monosemantically represents exactly the one feature it was initialized on. **(b)** Ordered k-hot, where expert one aligns with features 1–5 and each subsequent expert with the next five (same importance profile as (a)): the first expert monosemantically captures four of its five features. **(c)** Random k-hot, where each expert is assigned five random features so that experts share none but jointly cover all 20, with importance distributed randomly: each expert monosemantically represents only the most important feature(s) it was given. Across all three, the features an expert monosemantically represents are drawn from the features it was initialized to occupy.

occupies those features). The correlation holds under both Xavier and k-hot initialization (Table 1). With $E = 10$, a mean expert usage near $\sim 10\%$ reflects even load balancing across experts—the baseline against which the conditional usages should be read. The pattern is monotonic and large: in every case, activating an expert’s monosemantic feature(s) raises its usage well above the $\sim 10\%$ baseline, and when those feature(s) are the *only* active input the expert dominates routing. Under k-hot initialization the effect is complete—*100%* of the features an expert represents monosemantically are occupied by that same expert.

The table makes the specialization signature concrete. As an expert represents more features monosemantically, it becomes more specialized: its usage on arbitrary input stays near the $\sim 10\%$ balancing baseline, but conditional on its specialized features being active, its usage

Table 1: **Occupancy tracks monosemanticity, and usage rises sharply with it.** Monosemantic-feature and usage statistics per expert for $n = 100, m = 10, E = 10$, aggregated over 100 models per initialization scheme (1000 experts each for Xavier and k-hot). Experts are grouped by how many features they represent monosemantically. Columns report, for each group: the number of experts, mean unconditional usage (baseline $\sim 10\%$ for $E = 10$), mean usage when the expert’s monosemantic feature(s) are active, and mean usage when they are the *only* active input. Within each scheme, conditional usage climbs steeply with the number of monosemantic features—from 17.94% to 40.19% (Xavier) and 23.94% to 74.79% (k-hot) as the count grows—while the “only active” column saturates at 100%, and it does so for *every* k-hot group. Comparing schemes, k-hot yields both more monosemantic experts and stronger conditional routing than Xavier at every group.

Xavier Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage; feature(s) active (%)	Mean expert usage; only feature(s) active (%)
0	461	—	—	—
1	387	9.595	17.94	67.18
2	138	9.599	30.29	95.65
3	13	8.363	40.19	100.0
4	1	1.428	14.69	100.0
5	0	—	—	—
K-Hot Initialization				
Number of monosemantic features per expert	Number of experts (out of 1000)	Mean expert usage (%)	Mean expert usage feature(s) active (%)	Mean expert usage only feature(s) active (%)
0	335	—	—	—
1	382	10.00	23.94	100.0
2	227	10.02	46.61	100.0
3	47	10.09	62.00	100.0
4	8	9.95	70.30	100.0
5	1	9.62	74.79	100.0

jumps far above that of other experts—monotonically with the number of monosemantic features in both schemes. The single exception is the lone Xavier-initialized expert with four monosemantic features, whose unconditional utilization drops sharply to 1.428% (a single expert out of 1000); even so, when those four features are the only active input, that expert is still routed 100% of the time.

7 CONCLUSION

We investigated how experts affect superposition in MoEs and report three consistent, quantified findings. First, at matched parameters MoEs are systematically more monosemantic than dense networks: features-per-dimension falls monotonically toward the monosemantic limit of 1.0 as experts are added, while the dense model sits well above it. Second, the sharp phase changes that dense models reliably exhibit vanish in every MoE we test, replaced by smooth transitions that grow more monosemantic with network sparsity. Third, our feature-based definition of expert specialization is borne out in the usage data—experts organize around coherent features, with occupancy predicting relative monosemanticity by ~ 1 standard deviation under uniform importance and up to 100% of monosemantic features occupied by their expert under k-hot initialization. Crucially, this interpretability comes cheap: MoEs trail the dense model in loss by only ~ 0.03 – 0.08 , a gap that shrinks as experts are added. These findings are established in simple autoencoder toy models with synthetic data, and their generalization to large-scale transformers with unknown feature distributions remains open (Appendix A.6). Even so, the consistent result—comparable loss

with measurably more interpretable representations—directly challenges the prevalent view that mechanistic interpretability and model capability are fundamentally in tension. Future work should explore what favors monosemanticity in MoEs, how their training dynamics differ from dense models, and when specialization emerges—questions whose answers can inform the design of more interpretable, high-performing language models.

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A APPENDIX

A.1 MEASURING LOSS FOR VARYING SPARSITY & EXPERTS

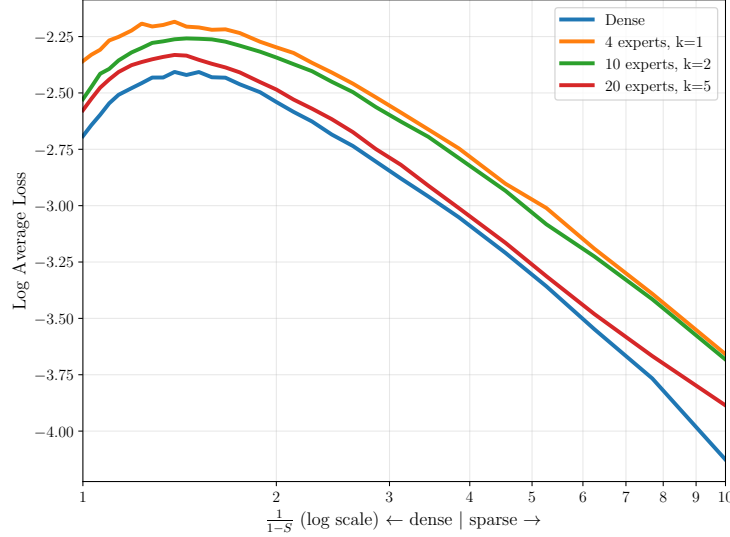


Figure 6: **The performance gap is small and closes with more experts.** Log average loss versus feature density ($\frac{1}{1-S}$) for dense ($m = 20$), MoE (4 experts, $k = 1$, $m = 5$), MoE (10 experts, $k = 2$, $m = 2$), and MoE (20 experts, $k = 5$, $m = 1$), all with uniform feature importance ($I_i = 1.0$), $n = 100$ input features, and averaged over five runs per sparsity level. The dense model attains the lowest loss at every sparsity level, but its margin over the MoEs is only ~ 0.03 – 0.08 and narrows monotonically as the expert count rises—so the most monosemantic MoEs (cf. Figure 3) are also the closest in loss.

A.2 THEORETICAL INTUITION FOR SUPERPOSITION

This section gives the mechanism behind the measured gap. In dense models, superposition emerges to exploit the gap between sparse features and dense computation, compressing rarely active features into shared dimensions. MoEs “eat this same gap” structurally through conditional computation: by aligning activation sparsity with feature sparsity, they remove the computational penalty for having dedicated, rarely-activating neurons. As Elhage et al. (2022) note, once a model only expends computation on active features, splitting polysemantic neurons into dedicated monosemantic ones becomes optimal—trading the *compression* of superposition for the *selection* of routing.

Geometrically, this shows up as reduced *interference*, which is exactly what the block-diagonal $W^\top W$ structure in Figure 2 and the below-dense features-per-dimension in Figure 3 measure. The global ratio of features represented to parameters stays constant, but the router partitions the feature space so that a feature routed to expert e competes for capacity only with the other features assigned to e . Interference is therefore governed by the local $(W^e)^\top W^e$ rather than the global $W^\top W$ of a dense model. This partitioning sharply lowers the number of interfering features per feature vector, relaxing the optimization pressure toward superposition and letting experts learn monosemantic representations.

A.3 ROUTER SUBSPACES ARE CONVEX CONES

In the regime of $k = 1$, the router function $g(x) = \text{softmax}(W^\top x)$ is equivalent to $\text{argmax}(W^\top x)$. The region routed to expert i can be represented as $\forall j \neq i, (w_i - w_j)^\top x > 0$ where w_i and w_j are row vectors of $W^\top$. This is a homogeneous linear inequality. Regions bounded by such inequalities are by definition convex cones. If a particular x satisfies this

inequality, then multiplying both sides by any positive scalar s will still satisfy the inequality. Furthermore, if x_1 and x_2 satisfy this inequality, then any $x = x_1\lambda + (1 - \lambda)x_2$ for $\lambda \in [0, 1]$ will also satisfy the inequality.

In the case of $k > 1$, the region of inputs which get sent to a particular expert e becomes a union of convex cones. Generally, the union of a convex cone is not itself a convex cone. Therefore, the understanding of experts occupying feature directions may not hold beyond $k = 1$.

A.4 EXPERT ROUTING WITH DIFFERENT INITIALIZATION SCHEMES

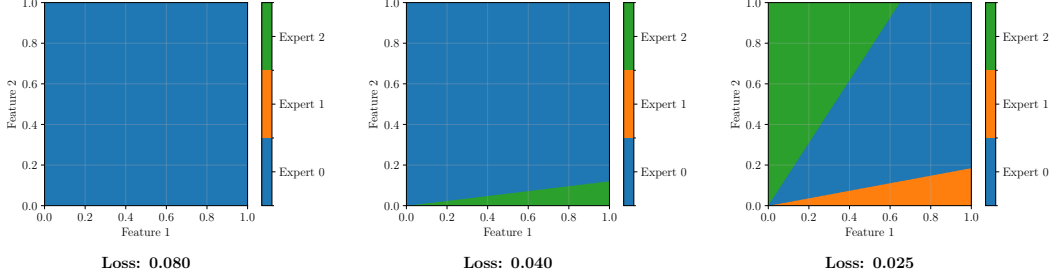


Figure 7: **Wider input-space distribution tracks lower loss.** Expert routing of three identical models ($n = 2$, $E = 3$, $m = 1$) that differ only in initialization. Loss falls monotonically as the input space is spread across more experts: the left model routes all inputs to a single expert (loss 0.08); the middle model peels off a small slice—inputs with feature 1 active—to a second expert (loss 0.04, a $2\times$ improvement); and the right model distributes across all three experts, one each for feature-1-only, feature-2-only, and both-active inputs (loss 0.025, the lowest and roughly $3\times$ better than the single-expert model).

In small models ($n = 2, m = 1, E > 1$), sweeping $E \in [2, 7]$, we consistently find that models distributing the input space across more experts achieve lower loss. Holding $n = 2$ lets us visualize directly which portions of the input space route to which experts (Figure 7), making the link between broader distribution and lower loss visible rather than merely inferred.

A.5 ANALYTIC MODEL EQUIVARIANCE

For the toy setup of single-layer, single-nonlinearity, top- $k = 1$ MoEs, there exists an exact theoretical map from any dense model to a monosemantic MoE that represents an equivalent number of active features under a sparsity constraint—the strongest possible form of the empirical trend reported in the main text, where superposition falls toward zero as experts are added.

Assume there exists an upper bound for the number of active features a for any input such that $\forall x \in D : |\{i : x_i \neq 0\}| \leq a$. Furthermore, assume that a is no greater than the hidden dimensionality, m , of an expert, providing an upper bound on the number of features a model has to represent. Assume also that the hidden dimensions is smaller than the total number of input features n ($a \leq m \leq n$). To construct the monosemantic MoE, for each possible subset $S \subseteq \{1, 2, \dots, n\}$ with $|S| \leq a$ —meaning the size of the subset of active features is smaller than or equal to a —create an expert which monosemantically preserves those features. (In fact, you can take only the subsets such that $|S| = a$.) The router then selects the expert which corresponds to those active features (of which there will never be more than a , by assumption):

$$\text{Router}(x) = \arg \max_S \mathbb{I}[\text{support}(x) = S]$$

where $\text{support}(x) = \{i : x_i \neq 0\}$. Since $|S| \leq a \leq m$, each expert has sufficient capacity to represent its assigned features without superposition. To reiterate, only a features are

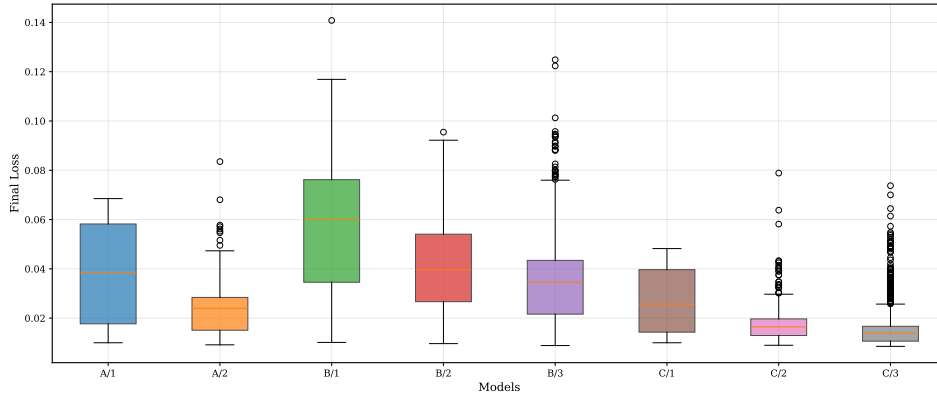


Figure 8: **Network sparsity lowers mean loss but raises its variance.** Model X/E reuses the architectures X of Figure 4, with E the total number of experts (i.e. network sparsity). Increasing network sparsity reduces mean loss while increasing localized variance, the effect sharpening as the expert count approaches the input feature dimensions. We attribute the variance to the relatively unstable training of MoEs versus dense models—present even though we train ten models per cell and keep the lowest-loss one.

active and every unique combination of active features receives its own dedicated expert with sufficient capacity to represent those features monosemantically. So, the number of possible experts needed is $\binom{n}{m}$.

The reconstruction for this theoretical MoE has zero loss only as the sparsity constraint holds (or goes to one in these toy models) because there is the chance more than m features could be active at one time ($a \not\leq m$), which would exceed the monosemantic representational capacity of the network (but the dense polysemantic could do no better unless features are correlated in the distribution). Therefore, even if $a \not\leq m$ sometimes, the polysemantic model encounters the same problem and the monosemantic MoE under this construction may still outperform it under looser sparsity constraints.

Thus, for any dense model, $f_{\text{dense}}(x) = \text{ReLU}(Wx + b)$ under the sparsity constraint $|\text{support}(x)| \leq a$, there exists a MoE model $f_{\text{MoE}}(x)$ such that $f_{\text{dense}}(x) = f_{\text{MoE}}(x)$ for all valid inputs. In the toy settings described in this paper, the sparsity constraint holds in the limit where sparsity goes to one. In practice, however, there is likely an upper bound on how many features a given amount of information can semantically encode, reflected in the size of meaningful embeddings of that data. A MoE with enough experts and only a tractable—hence interpretable—amount of superposition may therefore suffice to encode every feature present, which is the practical version of the near-monosemantic behavior we measure empirically.

A.6 LIMITATIONS

Our results should be read in light of several limitations. First, all experiments are conducted in controlled toy-model settings derived from superposition studies in sparse autoencoders (Elhage et al., 2022). This controlled design is precisely what enables the exact, parameter-matched measurements of interference and monosemanticity reported above, but it abstracts away much of the complexity of large-scale transformers—multiple layers of computation, attention, heterogeneous feature distributions, and realistic routing dynamics. The transferability of our quantitative findings to real-world MoE architectures therefore remains uncertain.

Second, our architectural and routing choices are intentionally restricted: simple experts, fixed top- k routing (often $k = 1$), equal parameter budgets between dense and MoE models, and no auxiliary load-balancing losses. These restrictions are what make the dense-versus-MoE comparison clean and attributable to network sparsity alone, but practical MoEs employ richer routing, variable expert capacities, and multi-task objectives (Lepikhin et al.,

2021; Fedus et al., 2022)—or move beyond reconstruction to next-token prediction—any of which could shift the representational behavior we measure.

Third, the feature distributions in our synthetic tasks—sparsity patterns, feature importance, and independence assumptions—are markedly simpler than those in natural data, so the exact numerical margins we report should be taken as characteristic of this controlled regime rather than as forecasts for natural data.

Finally, although we measure increased monosemanticity and reduced superposition in MoEs under fixed conditions—together with the small ~ 0.03 – 0.08 loss gap—we do not evaluate downstream performance trade-offs, training stability, or specialization dynamics at scale. Prior work suggests such dynamics can shift substantially with model size, optimization regime, and data diversity (Krajewski et al., 2024), so the favorable interpretability-versus-loss trade-off we report should be validated in that regime rather than assumed to hold.

Overall, our study provides mechanistic insights under clean experimental conditions, and the patterns we report—more experts, less superposition, at a small and shrinking loss cost—are consistent across every configuration we test. Validating them in large, realistic MoE systems is the clear next step; our current methods do not scale naturally to such models, which we regard as a primary direction for future work.